Week 3 – Day 1

Day 1 – What is Machine Learning?

Learn

Study:

- What is Artificial Intelligence?
- What is Machine Learning?
- What is Deep Learning?
- Traditional programming vs Machine Learning
- Supervised Learning
- Unsupervised Learning
- Classification
- Regression
- Features
- Labels

Important concepts

Understand this example:

Suppose we want to predict whether a motor will fail.

We might have:

Temperature
Voltage
Current
RPM
Vibration

These are our **features**.

And:

Motor Status

could be our **label**.

For example:

Temperature | Current | RPM | Vibration | Status

60	8.2	1480	0.15	Healthy
78	10.4	1320	0.42	Fault

The model tries to learn the relationship between the sensor measurements and the motor status.

Practical Work

Program 1 – Simple Classification Example

Use a small dataset that contains:

Hours Studied
Attendance
Exam Result

Example:

Hours Studied	Attendance	Result
2	60	Fail
4	70	Fail
6	80	Pass
8	90	Pass

Think about:

- What are the features?
- What is the label?
- Is this classification or regression?

Write your answers in:

ml_notes.txt

Program 2 – Explore a Dataset

Use the Iris dataset from scikit-learn.

Learn how to:

- Load the dataset.
- Display the data.
- Identify the features.
- Identify the target.
- Find the number of records.

Install

pip install scikit-learn

Git Task

Commit and push today's work.

Commit:

Day 1 - Introduction to Machine Learning